

DIABETES MANAGEMENT GUIDELINES

For most non-pregnant adults with type 2 diabetes, glycemic control is generally targeted at an HbA1c below 7.0%, though this target may be relaxed for older adults or those with significant comorbidities. Metformin remains a first-line therapy for type 2 diabetes due to its efficacy and low cost. However, Metformin is contraindicated in patients with an eGFR below 30 mL/min/1.73m², because reduced renal clearance increases the risk of metformin accumulation and lactic acidosis, a rare but serious adverse event. In patients with eGFR between 30 and 45, a reduced dose is typically recommended along with closer monitoring of renal function.

HEART FAILURE MANAGEMENT GUIDELINES

Patients with congestive heart failure require close monitoring of renal function and serum electrolytes, particularly when treated with loop diuretics such as furosemide, which can cause electrolyte disturbances and changes in kidney function. Heart failure patients with comorbid diabetes are at elevated risk for adverse cardiovascular and renal outcomes, and glycemic management in this population should account for renal function when selecting medications. Regular follow-up, generally at intervals no longer than 30 to 60 days for patients with more advanced heart failure, is recommended to detect early signs of decompensation.

CHRONIC KIDNEY DISEASE MANAGEMENT GUIDELINES

Medication dosing for patients with chronic kidney disease must be adjusted according to estimated glomerular filtration rate, as many medications rely on renal clearance and can accumulate to toxic levels when kidney function is impaired. Stage 4 chronic kidney disease, defined as an eGFR between 15 and 29 mL/min/1.73m², requires particular caution with renally cleared medications, including metformin and digoxin. Digoxin has a narrow therapeutic window and requires dose reduction and more frequent level monitoring as renal function declines, since reduced clearance increases the risk of digoxin toxicity. Patients with stage 4 chronic kidney disease should generally be seen no less frequently than every 90 days, given the risk of rapid clinical deterioration.

HYPERTENSION MANAGEMENT GUIDELINES

For most adults under 60 without significant comorbidities, a target blood pressure below 140/90 mmHg is generally recommended. ACE inhibitors such as lisinopril are commonly used as first-line therapy, particularly in patients with diabetes or proteinuria, though renal function and potassium levels should be monitored periodically during treatment.

ASTHMA MANAGEMENT GUIDELINES

Reliance on a reliever inhaler, such as albuterol, more than twice per week outside of exercise-related use generally indicates inadequate asthma control and should prompt consideration of step-up therapy, such as the addition of an inhaled corticosteroid. Patients with well-controlled asthma and infrequent reliever use typically do not require escalation of therapy.